

SMART INDIA HACKATHON 2026 — OFFICIAL PRELIMS

PITCH GUIDE

PROJECT A.U.R.A. | TEAM DELTA | Panimalar Engineering College (Autonomous)

Panel Head: Dr. S. Balaji (Professor, Dept of CSE)	Team Name: TEAM DELTA (Entry ID: SIH1566)
Date & Time: 7.9.2026, 6:00 PM – 8:00 PM	Designated Speakers: Kingston J & Nithyashree V
Platform: Google Meet (gad-jeft-gtj)	Pitch Structure: 5 Mins Presentation + 3 Mins Q&A;

■ VERIFIED SENSOR-BY-SENSOR PHYSICAL CAPABILITIES & DEPTH MATRIX

Sensor Modality	Hardware Model	Tested Range / Depth	Medium Penetrated	Target Detected	Trigger Math / Criteria
Microwave Radar	RCWL-0516 (3.18 GHz Doppler)	Depth: 0.5m – 2.5m (Open air: up to 6.0m)	Concrete rubble, brick, drywall, wood, plaster	Unconscious chest flutter, breathing, micro-twitching	Doppler frequency phase shift → Digital HIGH
Seismic Transducer	Piezoelectric Disc (12-bit ADC, Pin 35)	Pipes/Steel: 3m – 8m (Loose soil: 1.0m – 2.5m)	Solid concrete slabs, steel pillars, pipes, rebar	Rhythmic survivor tapping (knocks with stones/metal)	Impulse > 1.5 mV after baseline subtraction (12x gain)
Acoustic Mic	MAX9814 Low-Noise AGC Electret (Pin 32)	Acoustic: 2.0m – 5.0m (along void fissures)	Air crevices, collapse voids, vent/drainage ducts	Faint whispers, shouts, cries, gasps for air	Peak-to-peak amplitude > 8 units above dynamic noise floor
Bio-Scent VOC	MQ-135 Chemical Array (Pin 34)	Diffusion: 1.0m – 2.5m (confined voids)	Enclosed air pockets, collapse voids, crawlspaces	Exhaled CO2, ammonia (NH3), sweat VOCs, acetone	humanScentPPM > 4.5 PPM = (rawGas - 150) * 0.18
Δ-Jerk Vector	MPU-6050 6-Axis IMU (I2C 0x68)	Local: 0m – 1.5m (chassis contact ground)	Debris ground strata, probe structural chassis	Structural shifting, void settling, secondary collapse	Vector sqrt(dAx^2 + dAy^2 + dAz^2) > 0.4 m/s^2

Zone Color	Estimated Depth	Physical Meaning	Tactical Rescue Action Required
■ GREEN ZONE Surface Access	< 0.5 meters	High energy signature detected. Survivor is located directly under surface rubble.	Direct manual excavation using hand tools (crowbars, shovels). Immediate access.
■ RED ZONE Mid-Debris Core	0.5m – 2.5 meters	Moderate signal attenuation. Victim trapped beneath reinforced structural slabs.	Requires pneumatic rescue jacks, concrete chain cutters, and spreaders.
■ WHITE ZONE Deep Void	> 2.5m (up to 5m–6m)	Faint micro-impulses in basement cavities, deep pancake layers, or sewer voids.	Heavy structural shoring required. Directional core drilling to deploy oxygen tubes.

■ What Project A.U.R.A. CAN Penetrate: - Reinforced concrete slabs up to 20–30 cm thick (via microwave & seismic wave transmission). - Hollow brick masonry, sheetrock, plaster, and dry timber up to 1.5 – 2.0 meters thick. - Open-air crevices and complex debris void labyrinths up to 6.0 meters deep.	■ Physical Boundaries (What CANNOT be penetrated): - Solid sheet-metal barriers or thick armored steel plates (microwaves reflect completely). - Waterlogged, saturated mud or loose wet sand (seismic waves damp out within 1 meter). <i>(Honest trade-offs: We switch to seismic rebar contact when microwave encounters metal).</i>
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■■ THE 5-MINUTE WORD-FOR-WORD PRESENTATION SCRIPT

(Simple, compelling storytelling that anyone from an 8-year-old child to a 50-year-old professor can instantly grasp)

■■ [0:00 – 1:15] SPEAKER 1: KINGSTON J — The Story, The Human Crisis & The Problem

(Kingston looks directly into camera, smiling and speaking with calm, storytelling warmth)

"Good evening respected Panel Head Dr. Balaji Sir and honorable judges. We are **Team DELTA** from Panimalar Engineering College.

Let me tell you a story that happens during every major earthquake and landslide — like Wayanad or Turkey.

Imagine a multi-story building suddenly collapses into a pancake pile of concrete. Trapped underneath tons of rubble is a human being. They have water and air for only **72 hours — the Golden Window**. They can tap on a pipe with a small stone, or whisper for help.

Outside, our brave NDRF and SDRF rescue workers arrive. But they face a terrifying wall of silence. Flying drones with cameras cannot see through concrete. Mobile phones and walkie-talkies are completely dead under steel rebar. And foreign military radars weigh 20 kilograms, cost a massive **15 Lakh Rupees**, and can only penetrate 3 to 4 meters under ideal conditions.

Because rescue teams cannot see underneath, heavy excavators start digging blindly. That blind digging shakes the rubble, collapses the air pockets, and accidentally crushes the trapped victims.

We asked ourselves: *Why should someone die in the dark just because our sensors cannot see through concrete?* That is why Team DELTA built **Project A.U.R.A.** — the Autonomous Universal Rescue & Analysis Node."

■ [1:15 – 2:30] SPEAKER 2: NITHYASHREE V — How A.U.R.A. Works & What It Does

(Nithyashree speaks clearly and enthusiastically, holding up or pointing to the physical hardware)

"Thank you, Kingston. Respected judges, think of Project A.U.R.A. as a doctor's stethoscope, an acoustic ear, a radar eye, and a trained search dog — all combined into one compact handheld device.

Instead of guessing, A.U.R.A. uses four different physical senses running on an ESP32 dual-core computer:

- 1. Our Seismic Geophone on Pin 35:** It touches the concrete. When a survivor taps with a stone or pipe, it captures those shockwaves traveling through rebar and steel up to **8 meters away**.
- 2. Our Acoustic Ear on Pin 32:** It listens to the cracks in the rubble 500 times every second, separating faint whispers and cries from the noisy background generators.
- 3. Our 3.18 GHz Microwave Doppler Radar on Pin 33:** This is our superpower. While sound bounces around, radio microwaves pass through concrete up to **2.5 meters deep** to detect the tiny chest rise-and-fall of an unconscious breathing person.
- 4. Our MQ-135 Bio-Scent Gas Sensor on Pin 34:** Just like a rescue dog catches a scent, it sniffs out human exhaled breath and ammonia trapped in air voids, while warning rescuers if dangerous gas is leaking.

All of this is displayed on an onboard 8-slide mini-screen right on the probe, so a firefighter in heavy gloves can read the depth without needing a laptop."

■ [2:30 – 3:45] SPEAKER 1: KINGSTON J — How the Algorithm Works & The 3 Color Zones

(Kingston shares screen to show the live web dashboard with real-time green/red signals)

"Now, how does A.U.R.A. calculate where to dig?

Our device doesn't guess randomly. It uses a physics law called energy attenuation — the further sound travels through concrete, the weaker it gets. By measuring the exact wave speed between our microphone and geophone, A.U.R.A. calculates target depth using the formula: **Distance = 12 divided by square root of max energy**.

It then gives the rescue team Three Simple Color Zones on the screen:

- **GREEN ZONE (Under 0.5 meters):** The survivor is right near the surface. Responders can dig immediately with hand shovels.
- **RED ZONE (0.5 to 2.5 meters):** The victim is under heavy slabs. Teams must bring hydraulic jacks and slab cutters.
- **WHITE ZONE (Over 2.5 meters, up to 6 meters):** The victim is deep down in a basement cavity. Responders must shore up the walls and drill a narrow oxygen pipe first.

And here is our proudest engineering achievement: Every single number on this screen is **100% genuine physical sensor data**. If I tap the table right now, the needle spikes. If the room is quiet, it drops to silence. That real data is fed to Google Gemini Flash AI, which tells the commander: *'Survivor located 2.4 meters deep at 320 degrees North-Northwest.'*"

■ [3:45 – 4:45] SPEAKER 2: NITHYASHREE V — Cost, Honesty & The Big Mission

(Nithyashree delivers the economics and engineering honesty with authority)

"Now, let us talk about Cost and Feasibility.

A foreign search system like Delsar costs **12 to 15 Lakh Rupees**. Because of that price tag, only a few elite national teams have it. Your local district fire station has zero.

Project A.U.R.A. is built at **under ₹3,500 Rupees — less than 42 Dollars**. That means for the price of ONE imported machine, our government can equip **400 fire stations and rescue squads** across every district in India.

We also believe in honest engineering: Can A.U.R.A. penetrate everything? No. If there is a solid sheet of thick steel, microwaves reflect back. And in waterlogged, flooded mud, seismic waves damp out. But by combining radar, sound, tapping, and scent together, where one sensor faces a limit, the other three take over.

Our Phase 2 is already tested: adding LoRa mesh swarm networking so ten A.U.R.A. nodes can cover an entire football-field disaster zone simultaneously."

■ [4:45 – 5:00] BOTH TOGETHER — The Unforgettable Conclusion

KINGSTON: "Project A.U.R.A. is built to hear the unheard and bring hope into the darkest rubble."

NITHYASHREE: "Representing Panimalar Engineering College, we are **Team DELTA**, and our working prototype is live and ready for your test. Thank you!"

■ 3-MINUTE JUDGES' Q&A; SURVIVAL GUIDE

Q1: How can you prove your readings are real and not fake or simulated?

Kingston: 'Sir, our prototype is streaming right now. If I tap the desk, the seismic envelope spikes, the tap counter increments, and the decibel reading jumps. When I stop, the acoustic spectrum immediately reads SILENCE / NOISE FLOOR and victims show 0. Everything is wired directly to ESP32 ADC1 registers (Pins 32, 34, 35) with zero synthetic floors.'

Q2: Why does your microwave radar penetrate concrete when optical drones fail?

Nithyashree: 'Optical cameras use visible light with nanometer wavelengths that get absorbed by concrete surface atoms. Our RCWL-0516 uses 3.18 GHz microwave radio waves. These long radio waves travel right through non-metallic dielectric materials like concrete, brick, and wood, reflecting only when they hit water-rich human tissue and moving chest muscle.'

Q3: What if there is no Wi-Fi, 5G, or internet in a collapsed disaster area?

Kingston: 'A.U.R.A. is 100% offline-first. The ESP32 hosts its own Wi-Fi hotspot and local web server. Responders connect their tablet directly to the node without internet. Even if all tablets are broken, our built-in 8-slide OLED carousel displays depth, tap count, and gas levels right on the physical hardware.'

Q4: How did you bring the price down to under Rs. 3,500?

Nithyashree: 'Foreign systems use proprietary military FPGA chips. We replaced expensive hardware with smart software — running our 500Hz DSP on Core 1 of a mass-produced 240MHz ESP32-WROOM-32. Using commercial off-the-shelf sensors keeps our total component bill to exactly Rs. 3,420 INR.'